

Project Initialization and Planning Phase

Date	08/07/2026
Team ID	
Project Title	HDI Predictor - Human Development Index Prediction & Recommendation Platform
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

This proposal outlines an interactive, ML-powered web application that predicts a country's or hypothetical scenario's Human Development Index (HDI) tier from three core indicators, and recommends targeted improvements — replacing today's static, once-a-year HDI reporting with a live, explorable, decision-support tool.

Project Overview	
Objective	To develop an interactive, web-based Human Development Index (HDI) prediction and recommendation platform that classifies a country or scenario into a development tier — Very High, High, Medium, or Low — using three core indicators (Life Expectancy, Education, and GNI per capita), and to guide users on which indicator to improve to raise that tier.
Scope	The project covers the full pipeline: data collection from UNDP/World Bank sources, training a supervised ML classification/regression model on historical HDI data, building an interactive web frontend with real-time "what-if" simulation, and a backend inference API. It targets policymakers, researchers, students, and citizens who currently rely on static, once-a-year HDI reports.
Problem Statement	
Description	HDI is officially published only once a year by the UNDP as a static report. There is no widely accessible, interactive tool that lets a user enter their own values for life expectancy, education, and income and instantly see the resulting development tier, nor one that recommends which specific indicator to improve and by how much.
Impact	Without such a tool, policymakers cannot quickly explore intervention scenarios before committing budget, researchers must manually recompute HDI in spreadsheets, and students/citizens have no engaging way to understand how the three indicators interact. Solving this improves evidence-based policy planning, speeds up research workflows, and makes development data far more accessible and actionable.
Proposed Solution	
Approach	Train a machine learning model (classification for tier + regression for the underlying HDI score) on historical country-level HDI data, then expose it through a lightweight backend API. Build an interactive frontend where users adjust the three indicators and immediately see the predicted tier, score, and improvement guidance — turning a static annual statistic into a live, explorable simulation.
Key Features	- Real-time HDI tier & score prediction from 3 user-entered indicators. - Interactive "what-if" sliders that recompute the prediction live as values change.

	- Automated recommendation engine that identifies the weakest indicator and suggests a target improvement to reach the next tier. - Color-coded visual tier gauge (Very High/High/Medium/Low) and comparison view across multiple countries or scenarios. - Downloadable PDF/CSV report of any prediction, and optional login to save prediction history.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard CPU instance (no GPU required — models are lightweight regression/classification, e.g. Random Forest / Gradient Boosting)
Memory	RAM specifications	4–8 GB
Storage	Disk space for data, models, and logs	10 GB SSD
Software		
Frameworks	Backend framework for the prediction API	Flask / FastAPI
Frontend	Interactive UI framework	React.js / HTML-CSS-JS with Chart.js for live gauges & sliders
Libraries	ML & data libraries	scikit-learn, pandas, numpy, joblib, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, VS Code / PyCharm
Data		
Data	Source, size, format	UNDP Human Development Report Open Data (HDI, life expectancy, schooling, GNI per capita by country, ~190 records, csv); World Bank Open Data (supplementary indicators, csv)